CropClock — solution architecture

Decides which agricultural feeder is energised, and when, from crop water demand, the rain forecast and the feeder's own solar output.

1 Data in

free, no account

Sentinel-2 L2A — live data

Element84 Earth Search (AWS)
10 m red + NIR + scene class
every ~5 days

Weather archive — live data

Open-Meteo ERA5: observed rain,
FAO-56 ET_0, hourly irradiance

Forecasts as issued — live data

rain 1–2 days ahead,
irradiance day ahead
(Open-Meteo previous-runs)

Feeder register — planned

command area, pumps, MSKVY
solar MW, substation limit
prototype: 8 × 500 ha, approx.

2 Crop water layer

per feeder, daily

Cloud mask → NDVI — built

drop cloud, shadow, cirrus pixels
mean NDVI per feeder, each pass
28 clear passes in rabi 2025-26

Crop coefficient — built

Kc = 1.457 · NDVI – 0.1725
interpolated daily between passes
monsoon cloud → FAO-56 stages

Soil water balance — built

deficit += Kc · ET_0 – 0.8 · rain
irrigate at 80 mm (50% of TAW)
FAO-56, per feeder

rain + sun forecasts · feeder limits

3 Decision engine

re-plans every day

Rain-skip — built

≥15 mm forecast in next 48 h → defer
never past 75% of soil capacity
scored against rain that fell

Feeder scheduler (greedy) — built

need kWh = deficit ÷ 0.6 efficiency
1 sunniest forecast hours first
2 most water-stressed feeders first
3 ≤ 4 feeders on (substation cap)
4 every feeder keeps 8 h/day
5 overflow → hours nearest daylight

24 h × feeder switching plan

published 48 h ahead

4 Delivery

DISCOM switching — planned

plan sent to substation /
RDSS smart-feeder SCADA

Farmer notice — mocked

SMS / IVR in Marathi:
"your feeder runs 09–17"

Dashboard — built

deficit map, schedule vs rota,
savings — live on GitHub Pages

Backtest: 8 feeders, rabi 2025-26, real satellite passes, weather and forecasts — vs the MSEDCL day/night rota

–24.8%	**₹49.7 lakh**	**50.6% → 15.2%**	**41.9% → 56.3%**
grid energy (663 MWh)	subsidy avoided, one season	pumping at night	pump energy from solar

built in the prototype · mocked message designed, no gateway · planned needs DISCOM integration · live data public source

github.com/rahul428-2/cropclock